

Adesoye Oyeyiola

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EDUCATION

University of Northern British Columbia Prince George, BC
Bachelor of Science in Computer Science GPA: 4.14/4.33 | *Dean's List — Winter 2026* *Expected May 2028*

EXPERIENCE

Student Research Developer — Team Lead Jan 2026 – Present
Centre for Teaching, Learning and Technology (CTLT), UNBC *Prince George, BC*

- Leading backend development of an AI-assisted Teaching Dossier platform for 50+ educators; running sprint planning, code reviews, and weekly stakeholder updates to faculty
- Built a 5-layer REST API (34 endpoints) in Python/FastAPI with JWT auth, Argon2, rate limiting, and RBAC; designed PostgreSQL schema with year-versioned educator profiles and 16+ Alembic migrations
- Integrated backend with ASP.NET Core MVC frontend; deployed to Railway with automated CI migrations
- Co-first-authored *An Architecture of an AI-Assisted Teaching Dossier Development* — submitted to ICCE 2026 (under review)

Data Engineer Intern Oct 2023 – Jan 2024
Muzedata *Virginia, USA (Remote)*

- Maintained Python ETL pipelines processing real-time crypto market data with 99.9% accuracy across 50+ trading pairs
- Improved pipeline performance by 40% through query optimization, indexing, and automated data quality checks with anomaly detection

RESEARCH & VOLUNTEER

Robotics Research Volunteer Jan 2026 – Present
Human-Robot Interaction Lab, UNBC (Prof. Shruti Chandra) *Prince George, BC*

- Contributing to Blossom, an open-source social robot; writing servo calibration algorithms and building a real-time emotion recognition system using DeepFace/FER
- Co-first-authored *Guardians of the Planet* — accepted at ACM IDC 2026 (R&D Challenge Finalist, Brighton, UK); presenting at conference; prototyped multi-platform sustainability education system using Scratch, Kivicube AR, and physical robotics

PROJECTS

U-Course Advisor | *Python, FastAPI, PostgreSQL, pgvector, Claude, Auth0, Celery* 2026

- Built a full-stack AI course advising platform for UNBC; prerequisite validation engine supports AND/OR dependency trees and enforces minimum grade requirements across 400+ courses
- Built a RAG-powered AI chat advisor using Claude with multi-layer guardrails, routing queries between vector semantic search (Voyage AI embeddings via pgvector) and a degree-progress-aware recommendation engine
- Secured with Auth0 JWT auth, async SQLAlchemy, Celery background tasks, and Sentry monitoring; deployed on Railway with separated production/development environments

Teaching Experience Analytics Platform | *Python, FastAPI, PostgreSQL, SQLAlchemy, ASP.NET* 2025–26

- Full backend with 7 routers, JWT, Argon2, rate limiting, 16+ Alembic migrations, year-versioned educator profiles; integrated with ASP.NET Core MVC frontend and deployed on Railway

Huffman Compression Tool | *Java, Algorithms, Data Structures* 2025

- Lossless compression with custom binary format (HUF1 header), priority-queue Huffman tree, and bit-level I/O classes; handles edge cases including empty and single-symbol files

Blossom Robot — Adventures in Technology | *Python, pypot, Dynamixel XL-320* 2026

- Motor control sequences, calibration CLI, and live demo system for 50+ students at UNBC; contributed to open-source Cornell HRI robotics platform

TECHNICAL SKILLS

Languages: Python, Java, SQL, JavaScript, TypeScript, C/C++, C#
Backend & Data: FastAPI, SQLAlchemy, PostgreSQL, pgvector, Alembic, REST APIs, Celery
AI/ML: RAG, Claude, Voyage AI Embeddings, Vector Search
Auth & Security: JWT, Auth0, Argon2, OAuth, Role-Based Access Control
Tools: Git, Docker, Railway, Linux, Agile/Scrum, CI/CD, .NET, Sentry, OpenCV, ROS

CERTIFICATIONS & ACHIEVEMENTS

ML Specialization — Andrew Ng, DeepLearning.AI / Stanford (Coursera) 2026

Techvantage (Founder, 2022–2023): trained 141 teenagers in tech skills, raised \$2,200 CAD — Lagos, Nigeria